AshvaTrace

AI-Powered CNC Fault Investigation & Verification

AI Builder Cup 2026 · Manufacturing: Intelligent Operations & Industrial Efficiency

Team: AshvaTrace · October 2026

The Problem: Unplanned Downtime is Devouring Manufacturing

$1.4T	$124K	49→81 min	52.7%
Annual global downtime cost	Median cost per hour of downtime	MTTR increase (2019→2024)	Downtime reduction with AI (NIST)

When a CNC machine drifts off its expected power baseline, operators face a wall of sensor data. Finding the root cause takes 30-90 minutes of manual investigation — or worse, a wrong guess that sends a technician to the wrong part. Every minute of misdiagnosis is $2,000+ in lost production.

The gap: AI can propose hypotheses, but nobody trusts AI to verify the fix. You need a system where AI investigates, a human approves, and deterministic code proves the outcome.

AshvaTrace: AI Proposes, Code Verifies

A closed-loop fault investigation system where AI does the thinking and deterministic code does the proving.

1. DETECT
Real-time power/vibration/cycle-time monitoring with configurable thresholds. Alerts fire the moment a machine drifts.

2. INVESTIGATE (AI)
Gemini (Vertex AI) runs an agentic investigation: calls tools to pull real sensor data, inspects tool images via Vision AI, retrieves relevant SOPs, and generates ranked hypotheses with evidence.

3. APPROVE (Human)
Engineer reviews hypotheses, selects the most likely root cause, and approves a simulated intervention.

4. VERIFY (Code)
Deterministic code (NOT AI) checks actual metrics against predicted thresholds. Pass/fail is computed, not guessed.

Why AshvaTrace is Different

AI-Proposes / Code-Verifies Loop
Every intervention is verified by deterministic code, not another AI call. This is the trust boundary that makes the system production-ready.

Agentic Investigation with Visible Trace
The AI doesn't just output hypotheses — it shows its work: which tools it called, what data it saw, and how it reasoned. Engineers can audit the AI's thinking.

RAG over Plant SOPs
Hypotheses are grounded in the plant's actual standard operating procedures, not generic LLM knowledge. The AI cites specific SOP sections.

Vision AI Tool Inspection
Gemini analyzes synthetic tool inspection images to assess wear condition, adding a visual dimension to the investigation.

Cost Impact Dashboard
Quantifies the financial impact: downtime cost, MTTR vs. industry benchmark, energy waste, and estimated savings from faster resolution.

Honesty by Design
If the AI call fails, the system says so. It never substitutes canned content. Failed investigations are logged and the incident reverts to "open".

Google Cloud Architecture

Gemini on Vertex AI
Agentic investigation with function calling, multi-round tool use, and a two-pass propose/critique reasoning chain. Vision AI for tool inspection.

BigQuery
Persistent storage for incidents, alerts, SOPs, and audit logs. Enables historical analysis and cost-impact reporting.

Cloud Run
Serverless deployment of the full-stack app (React frontend + Express backend in a single container). Scales to zero, pay-per-request.

Pub/Sub
Event-driven alerting: threshold crossings publish to a Pub/Sub topic, enabling downstream consumers (dashboards, notifications, CMMS integration).

Cloud Build
CI/CD pipeline: git push → Docker build → Cloud Run deploy. Zero manual deployment steps.

React + Vite + TypeScript
Frontend with real-time SSE alerts, interactive charts, drag-and-drop dashboard, and investigation trace visualization.

Impact: What AshvaTrace Delivers

10 min
Target MTTR (vs. 81 min industry avg)

88%
Faster fault resolution

$112K
Saved per avoided downtime hour

4
AI capabilities in one loop

The cost-impact dashboard quantifies the value in real-time: every incident resolved faster saves $2,000+/min in production. By reducing MTTR from 81 to 10 minutes, AshvaTrace delivers an 88% improvement in fault resolution speed — directly translating to millions in annual savings for a mid-size plant.

Benchmark sources: ABB 2023 ($124K/hr downtime), Siemens 2024 (81 min MTTR), NIST 2023 (52.7% reduction with AI).

Live Demo: 3-Minute Walkthrough

0:00 Plant Overview — 6 machines, CNC-03 in CRITICAL state with power deviation +15.2%

0:30 Open Incident — AI investigation begins, watch the agentic trace: tool calls, data retrieval, reasoning

1:00 Hypotheses — 4 ranked causes with evidence for/against, SOP citations, and confidence scores

1:30 Vision AI — Run tool inspection, see Gemini analyze the tool image and report wear condition

2:00 Approve & Simulate — Engineer selects "Replace spindle bearing", AI predicts the outcome

2:30 Verify — Deterministic code checks actual metrics: PASS. Incident closed, audit log complete.

3:00 Cost Impact — Dashboard shows $112K saved, MTTR 10 min vs. 81 min benchmark

Beyond the Hackathon

Phase 1 (Now)
Full-stack app on Cloud Run, real Gemini investigation, BigQuery persistence, Pub/Sub alerting, Vision AI inspection, cost dashboard.

Phase 2 (3 months)
Connect to real PLC/CNC data via OPC-UA or MQTT. Replace synthetic data with live sensor streams. Add Dataflow for real-time processing.

Phase 3 (6 months)
Multi-plant deployment. CMMS integration (SAP PM, Maximo). Predictive maintenance: train Vertex AI models on historical incident data to predict failures before they happen.

Phase 4 (12 months)
Digital twin: what-if simulation of interventions before they happen. Autonomous mode: AI proposes AND verifies, human only approves. Fleet-level analytics across 100+ machines.

AshvaTrace is live at: https://ashvatrace-206380300251.us-central1.run.app
GitHub: github.com/akanthed/AshvaTrace